

## Market Basket Analysis Report

### Why This Analysis is Important

Market basket analysis (MBA) is a vital data mining technique to understand customer purchasing behaviors in retail and e-commerce industries. This analysis focuses on identifying patterns and relationships between products frequently bought together, providing valuable insights for cross-selling, product bundling, and inventory management. By applying MBA, businesses can:

- Increase sales by recommending relevant products.
- Optimize store layouts to place complementary items together.
- Enhance customer satisfaction through personalized marketing.
- Improve supply chain efficiency by forecasting demand for related products.

In essence, MBA is a foundational tool for driving customer-centric, data-driven decision-making.

### Rules Applied in the Analysis

Four key metrics were used to uncover relationships between products:

1. **Frequency:** This determines the number of transactions where two products (Product1 and Product2) are purchased together. For instance, "whole milk" and "other vegetables" appeared together in 1,458 transactions.
2. **Support:** The percentage of total transactions that include both Product1 and Product2. Calculated as:

$$\text{Support} = (\text{Frequency of products 1 \& 2} / \text{Total number of transactions}) * 100$$

Example: The support for "whole milk" and "other vegetables" is 3.76%, indicating that they appear together in 3.76% of all transactions.

3. **Confidence:** Measures the likelihood that Product2 is purchased when Product1 has already been bought. Calculated as:

$$\text{Confidence} = (\text{Frequency of both products} / \text{Frequency of the product on the LHS}) * 100$$

Example: The confidence for "whole milk" and "other vegetables" is 58.27%, meaning customers who bought "whole milk" are 58.27% likely to also buy "other vegetables."

4. **Lift:** Indicates how much more likely customers are to buy Product2 when Product1 is purchased compared to buying Product2 independently. Calculated as:

$$\text{Lift} = \text{Support (Both Products)} / (\text{Support(Product1)} * \text{Support(Product2)})$$

Example: The lift of 2.85 for "whole milk" and "other vegetables" suggests that buying "whole milk" increases the likelihood of buying "other vegetables" by 2.85 times compared to buying "other vegetables" independently.

### Top 10 Product Pairs

The analysis identified the following top 10 product combinations based on frequency:

| Rank | Product1   | Product2         | Frequency | Support (%) | Confidence (%) | Lift |
|------|------------|------------------|-----------|-------------|----------------|------|
| 1    | Whole Milk | Other Vegetables | 1,458     | 3.76        | 58.27          | 2.85 |
| 2    | Whole Milk | Rolls/Buns       | 1,312     | 3.38        | 52.44          | 2.32 |
| 3    | Whole Milk | Soda             | 1,074     | 2.77        | 42.93          | 1.68 |
| 4    | Yogurt     | Whole Milk       | 1,062     | 2.74        | 79.61          | 5.14 |
| 5    | Rolls/Buns | Other Vegetables | 997       | 2.57        | 58.1           | 2.84 |
| 6    | Soda       | Other Vegetables | 875       | 2.26        | 57.79          | 2.83 |
| 7    | Whole Milk | Tropical Fruit   | 786       | 2.03        | 31.41          | 0.84 |
| 8    | Whole Milk | Root Vegetables  | 780       | 2.01        | 31.18          | 0.86 |
| 9    | Yogurt     | Other Vegetables | 766       | 1.98        | 57.42          | 2.81 |
| 10   | Soda       | Rolls/Buns       | 748       | 1.93        | 49.41          | 2.19 |

### How This Analysis is Useful for Data-Driven Decision Making

This Market Basket Analysis provides actionable insights for business strategy, including:

1. **Product Recommendations:**

- Use high-lift product pairs (e.g., "Yogurt" and "Whole Milk") for cross-selling opportunities in online and in-store settings.

- Personalized marketing campaigns can recommend complementary items based on customer purchase history.

## **2. Product Placement and Bundling:**

- Place frequently paired products (e.g., "Whole Milk" and "Other Vegetables") close to each other in stores to encourage impulse purchases.
- Bundle products with high confidence scores to increase transaction value.

## **3. Promotional Strategies:**

- Offer discounts on pairs with high frequency but low confidence to encourage repeat purchases.
- Use lift as a guide to design promotions that amplify related product sales.

## **4. Inventory Management:**

- Ensure adequate stock of frequently paired products (e.g., "Whole Milk" and "Rolls/Buns") to avoid losing sales opportunities.
- Forecast demand more accurately by understanding product relationships.

By leveraging these insights, businesses can achieve higher customer satisfaction, improved sales performance, and better resource allocation.

## **Additional Rules Applied**

The following SQL rules and logic were applied to conduct this analysis:

### **1. Use of Common Table Expressions (CTE):**

- A CTE named Market was created to simplify the process of calculating metrics like Frequency, Support, Confidence, and Lift.

### **2. SQL Aggregation and Joining:**

- The COUNT function was used to calculate the frequency of product pairs.
- Self-joins were utilized to pair products purchased by the same customer.

### **3. Subqueries:**

- Subqueries were implemented to calculate the total transactions, the frequency of Product1 (LHS), and the frequency of Product2 (RHS).

### **4. Filtering and Sorting:**

- A WHERE clause ensured that each product pair is unique (using `GD1.itemDescription > GD2.itemDescription`).
- Results were sorted by frequency in descending order to identify the top 10 product pairs.

By applying these rules and techniques, the analysis was able to provide precise and meaningful results for actionable business strategies.